

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)*

Assessment Tool Weightage: 40%

QUESTIONS

- 1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.**

Answer:

Sampling determines the number of pixels used to represent an image. If the sampling rate is too low, fine details are lost, causing blur and aliasing. Increasing the sampling rate (higher spatial resolution) or using proper interpolation methods such as bilinear or bicubic interpolation improves image quality.

- 2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.**

Answer:

In low-light conditions, the image sensor receives fewer photons, resulting in dark and noisy images. The camera may increase ISO or exposure time, which can introduce additional noise or motion blur. Using a better sensor, proper illumination, or noise reduction techniques improves image quality.

- 3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.**

Answer:

Aliasing occurs when the sampling rate is lower than the required Nyquist rate, causing jagged edges or false patterns. Quantization is not the primary cause but may worsen visual quality if intensity levels are too low. Applying anti-aliasing filters before sampling and increasing the sampling rate reduces aliasing.

- 4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.**

Answer:

- **Pixel Resolution:** Determines the number of pixels in the image and affects spatial detail.
- **Intensity Resolution:** Determines the number of gray levels and affects brightness accuracy.

In medical imaging, high pixel resolution helps detect small tumors, while high

intensity resolution improves the visibility of subtle tissue differences.

5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.

Answer:

This involves the **High-Level Computer Vision** stage because the system interprets images, detects objects, tracks people, and makes decisions in real time. It combines image processing with object recognition and analysis.

6. Explain how image formation models help in improving image quality in applications such as autonomous driving.

Answer:

Image formation models describe how light interacts with objects and the camera. Understanding illumination, reflection, and camera geometry helps correct distortions, improve contrast, and enhance visibility, enabling accurate lane detection, obstacle recognition, and navigation.

7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Answer:

Quantization converts continuous intensity values into discrete gray levels. Higher quantization levels preserve more image detail, while lower levels reduce file size but may introduce banding and loss of fine details.

8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Answer:

Sensor noise occurs due to electronic interference, thermal effects, or low-light conditions during image acquisition. Noise can be minimized by using high-quality sensors, proper illumination, lower ISO settings, and applying image filters such as Gaussian or Median filters.

9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Answer:

1. **Low-Level Vision:** Captures and enhances the face image.
2. **Intermediate-Level Vision:** Extracts facial features such as eyes, nose, and mouth.
3. **High-Level Vision:** Compares extracted features with a database and identifies the person.

10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

Answer:

Low sampling produces fewer pixels, reducing image detail. Poor image formation due to improper focus or lighting also decreases clarity. Increasing image resolution and improving camera focus and illumination enhances object visibility.

11. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.

Answer:

CCD Sensor	CMOS Sensor
Higher image quality	Slightly lower image quality
Low noise	More noise (modern CMOS is improved)
Higher power consumption	Lower power consumption
Slower readout	Faster readout
Used in scientific imaging	Used in smartphones and surveillance

12. A digital image appears distorted due to improper image formation. Identify possible causes and suggest improvements based on image formation models.

Answer:

Possible causes include poor lens quality, incorrect focus, motion blur, or uneven lighting. Improvements include proper camera calibration, autofocus, high-quality lenses, and better illumination to produce clearer images.

13. Explain how the features of Computer Vision systems enable applications such as automated surveillance.

Answer:

Computer Vision provides features such as object detection, motion tracking, face recognition, behavior analysis, and event detection. These enable automated surveillance systems to monitor areas continuously and detect suspicious activities without human intervention.

14. A system processes satellite images for land analysis. Explain how digital image fundamentals and acquisition methods influence the accuracy of results.

Answer:

High-resolution image acquisition captures finer land details. Proper sampling, accurate sensing, and good intensity resolution improve feature extraction and classification accuracy for forests, roads, buildings, and agricultural land.

15. Given an application like OCR (Optical Character Recognition), explain how sampling, quantization, and image acquisition collectively affect performance.

Answer:

Good image acquisition captures clear text. High sampling resolution preserves character shapes, while sufficient quantization maintains contrast between text and background. Together, they improve OCR accuracy and reduce recognition errors.

16. A high-resolution image consumes excessive storage and processing time. Explain how sampling and quantization can be adjusted to optimize performance without significant quality loss.

Answer:

The sampling rate can be slightly reduced to lower image size while maintaining acceptable detail. Quantization can also be optimized by reducing unnecessary intensity levels, balancing image quality with storage and processing efficiency.

17. An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.

Answer:

Uneven lighting causes non-uniform illumination during image formation, reducing contrast in some regions. Techniques such as histogram equalization, adaptive contrast enhancement, and proper lighting improve image visibility.

18. In a medical imaging system, explain how improper quantization can lead to loss of critical diagnostic information.

Answer:

Low quantization levels reduce the number of intensity values, causing loss of subtle gray-level differences. Important features such as small tumors or tissue boundaries may become invisible, affecting diagnosis accuracy.

19. A self-driving car system relies on visual input. Explain how image acquisition and Computer Vision features contribute to reliable decision making.

Answer:

High-quality image acquisition provides clear images of roads and obstacles. Computer Vision algorithms detect lanes, traffic signs, pedestrians, and vehicles. Accurate feature extraction and object recognition enable safe navigation and real-time decision making.

20. An application requires detecting fine details in images. Explain how sampling rate and intensity resolution should be chosen to meet this requirement.

Answer:

A high sampling rate should be used to capture fine spatial details, while high intensity resolution (more gray levels or color depth) preserves subtle brightness differences. Together, they improve the detection of small objects and fine image features.

Code of Conduct:

*I **G. Gopi Chand (192424384)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

G. Gopi Chand